

Hearing

The living world is rarely silent. Waves of pressure travel through air, water, and the ground, carrying energy that animals feel as vibrations or hear as sound. Some insects have ears on their antennae, or even on their legs, but in mammals and other vertebrates, the real work of hearing happens once sound is channeled into the head. Here, sensory cells respond to its intensity and pitch (how high or low the sound is), triggering a stream of nerve signals that the brain interprets as sound. For many animals, hearing is essential for communication. Crucially, hearing can reveal the approach of predators, or the exact position of prey.



Internal cavity of left ear is smaller and lower than the right

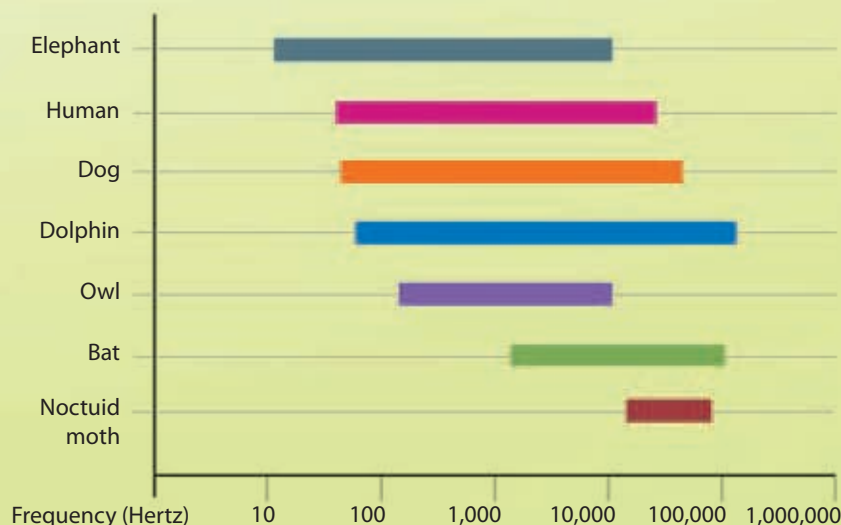
Skull of a Tengmalm's owl

Lopsided ears

Most birds' ears are hidden by the feathers that cover their heads. Despite this, many have extremely good hearing. Far from muffling the sound, the feathers around owls' ears form tunnels that collect and channel the sound like mammals' external ears do. A Tengmalm's owl can track prey using its ears alone. Its two ear cavities are in slightly different positions, which enables it to get an exact "fix" on its target.

Hearing range

A sound's pitch is measured in waves per second, or Hertz (Hz). Humans can sense sounds with a pitch of between 20 and 20,000 Hz, depending on age. Elephants can sense deeper sounds than we can, but many more animals can hear ultrasound, which is sound too high for us to hear. Ultrasonic hearing is important to animals that use echolocation to hunt.



Club-shaped antenna

Branching hair picks up airborne sound waves

All a-quiver

The antennae of insects are multipurpose sense organs. Fruit flies have antennae that sense air currents and gravity, but they also act as ears. Each antenna has a single branching hair. Sound waves make the hair vibrate, activating a hearing organ. When a male fruit fly courts a female, it vibrates one of its wings. The female senses this with her antennae and lets him mate.

Insect ears

Insects have ears in a range of different body parts. Despite this, their ears often work in a similar way. Many insects hear with tympanal organs, which are like eardrums on their body's surface. These often pick up sounds of a certain pitch, helping insects to dodge particular predators and to locate potential mates.

Mouthparts detect ultrasonic sounds



Hawk moth

Some hawk moths have ears in their mouthparts. They use them to avoid attack by hunting bats.

Slit-shaped eardrum on forewing



Lacewing

Lacewings use their wing-mounted ears, like hawk moths, to detect the sounds of bats approaching.

Eardrum below middle joint of foreleg



Cricket

Crickets have eardrums on their front legs. These let them hear courtship calls of other crickets.

Eardrums underneath front of thorax



Tachinid fly

Using eardrums on their thorax, these parasitic flies often track down other insects by their calls.